

JEE MAIN REVISION MASTER: ELECTROCHEMISTRY 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 35 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Faraday's Laws of Electrolysis	13 PYQs	Quantitative Electrolysis
High	Nernst Equation & Cell Potential	9 PYQs	Core Formula
High	Electrolytic Conductance & Cell Constant	9 PYQs	Conductance
High	Commercial Batteries, Fuel Cells & Corrosion	9 PYQs	Applied Electrochemistry
High	Electrode Potential & Delta G Relation	8 PYQs	Thermodynamics
Moderate	Kohlrausch's Law & Weak Electrolyte Dissociation	5 PYQs	Electrolyte Behavior

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Faraday's Laws of Electrolysis	$w = \frac{M}{n} \cdot I \cdot t$, $\text{Moles of metal} = \frac{I \cdot t}{n \cdot F}$, $F \approx 96500 \text{ C mol}^{-1}$
Nernst Equation & Cell Potential	$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log_{10} Q$, $\Delta G^{\circ} = -n F E^{\circ}_{\text{cell}} = -2.303 RT \log_{10} K$
Electrolytic Conductance & Cell Constant	$\kappa = \frac{1}{R} \left(\frac{l}{A} \right) = G \cdot G^{\circ}$, $\Lambda_m = \frac{1000 \cdot \kappa}{M}$ (κ in S cm^{-1})
Commercial Batteries, Fuel Cells & Corrosion	Lead Storage Discharge: $\text{Pb(s)} + \text{PbO}_2\text{(s)} + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4\text{(s)} + 2\text{H}_2\text{O}$, $\eta = \frac{\Delta G}{\Delta H}$
Electrode Potential & Delta G Relation	$\Delta G = -nFE$, $\Delta G^{\circ}_3 = \Delta G^{\circ}_1 + \Delta G^{\circ}_2$ $\implies n_3 E^{\circ}_3 = n_1 E^{\circ}_1 + n_2 E^{\circ}_2$
Kohlrausch's Law & Weak Electrolyte Dissociation	$\Lambda_m^{\circ} = \nu_+ \lambda_+^{\circ} + \nu_- \lambda_-^{\circ}$, $\alpha = \frac{\Lambda_m}{\Lambda_m^{\circ}}$, $K_a = \frac{c \alpha^2}{1 - \alpha}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Faraday's Laws of Electrolysis (High Yield — 13 PYQs)

Core Principle: First and second laws of quantitative electrolysis: Mass deposited $w = Z I t = (M \cdot I \cdot t / n F)$. Total charge passed $Q = I t = n_e F$.

■■ **Caution / JEE Pitfall:** 1. Using an incorrect n-factor (electrons transferred per molecule). E.g. for O_2 from H_2O , $n = 4$; for H_2 , $n = 2$. 2. Time t must strictly be substituted in seconds (s).

■ **Speed Trick:** Gas volume at STP: Moles $\times 22.4$ L. For electroplating, Volume = Area $\times$ Thickness and Mass = Density $\times$ Volume.

Representative Source PYQs:

- [2026 Shift] Q#4 (MCQ): "During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at c..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): "The correct statement/s about Galvanic cell is/are (a) Current flows from cathode to anode (b) Anode is positive terminal (c) If E_{c} ..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Match the following select the correct option for the quantity of electricity, in $Cmol^{-1}$ required to deposit various metals at cathode..." [Ans: N/A]

3.2 Nernst Equation & Cell Potential (High Yield — 9 PYQs)

Core Principle: Calculation of non-standard cell potential $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.0591 / n) \log_{10} Q$ at 298 K, equilibrium constant K_{eq} , and concentration cells where $E^{\circ}_{\text{cell}} = 0$.

■■ **Caution / JEE Pitfall:** 1. Forgetting to raise concentrations to their stoichiometric powers in reaction quotient Q . 2. Confusing oxidation vs reduction potential signs: $E^{\circ}_{\text{red}} = -E^{\circ}_{\text{ox}}$. 3. In concentration cells, anode is always the dilute compartment ($c_1 < c_2$).

■ **Speed Trick:** At 298 K, $E^{\circ} = (0.0591 / n) \log_{10} K$. For spontaneous cell operation, $E_{\text{cell}} > 0$ and $\Delta G < 0$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Given below are the half-cell reactions: $Mn^{2+} + 2e^- \rightarrow Mn$ ($E^{\circ} = -1.18V$) $2(\text{Mn}^{3+} + e^- \rightarrow \text{Mn}^{2+})$..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The correct statement/s about Galvanic cell is/are (a) Current flows from cathode to anode (b) Anode is positive terminal (c) If E_{c} ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "For a given half cell, $Al^{3+} + 3e^- \rightarrow Al$ on increasing of aluminium ion, the electrode potential will..." [Ans: N/A]

3.3 Electrolytic Conductance & Cell Constant (High Yield — 9 PYQs)

Core Principle: Specific conductivity $\kappa = (1/R) \left(l/A \right) = G \cdot G^*$. Molar conductivity $\Lambda_m = (1000\kappa / M)$ (with κ in $S\ cm^{-1}$) or $\Lambda_m = (\kappa / 1000M)$ (with κ in $S\ m^{-1}$).

■ **Caution / JEE Pitfall:** Unit confusion between cm and m. If κ is in $S\ m^{-1}$, use $\Lambda_m = (\kappa / c)$ (c in $mol\ m^{-3}$). If in $S\ cm^{-1}$, use $\Lambda_m = (1000\kappa / M)$ (M in $mol\ L^{-1}$).

■ **Speed Trick:** Cell constant $G^* = (l/A)$ is an intrinsic geometric property of the conductivity cell and is strictly independent of the electrolyte filled.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The conductivity of centimolar solution of KCl at 298 K is $0.021\ Ohm^{-1}cm^{-1}$ and the resistance of the cell containing ..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The electronic conductance depends on..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "The resistance of $0.1M$ weak acid HA in a conductivity cell is $2 \times 10^3\ Ohm$. The cell constant of the cell is $0.78\ cm^{-1}$ and ..." [Ans: N/A]

3.4 Commercial Batteries, Fuel Cells & Corrosion (High Yield — 9 PYQs)

Core Principle: Primary cells (dry cell, mercury cell), secondary lead-acid batteries (charge/discharge cycles), H_2 - O_2 fuel cells (thermodynamic efficiency $\eta = (\Delta G / \Delta H)$), and electrochemical corrosion prevention.

■ **Caution / JEE Pitfall:** In a lead storage battery, the density of H_2SO_4 decreases during discharge because sulfuric acid is consumed and water is produced.

■ **Speed Trick:** In galvanic cells: Anode is negative, Cathode is positive (An Ox, Red Cat). In electrolytic cells: Anode is positive, Cathode is negative.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Given below are the half-cell reactions: $Mn^{2+} + 2e^- \rightarrow Mn(E^0 = -1.18V)$ $\text{\textit{\textit{Mn}}}^{3+} + e^- \rightarrow \text{\textit{\textit{Mn}}}^{2+}$..." [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at c..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): "The correct statement/s about Galvanic cell is/are (a) Current flows from cathode to anode (b) Anode is positive terminal (c) If E_{cell} ..." [Ans: N/A]

3.5 Electrode Potential & Delta G Relation (High Yield — 8 PYQs)

Core Principle: Relationship between standard reduction potentials and Gibbs free energy: $\Delta G^\circ = -nFE^\circ$. When combining half-reactions, always add extensive ΔG values, never add intensive E° potentials directly!

■ **Caution / JEE Pitfall:** Directly adding E° potentials of two half-cells to obtain E° of a composite couple (e.g. $Fe^{3+} \rightarrow Fe$ from $Fe^{3+} \rightarrow Fe^{2+}$ and $Fe^{2+} \rightarrow Fe$). You MUST weight by electron transfer count: $n_3 E_{3^\circ} = n_1 E_{1^\circ} + n_2 E_{2^\circ}$!

■ **Speed Trick:** Always write $\Delta G = -nFE$ before combining half-reactions. A more positive standard reduction potential E° denotes a stronger oxidizing agent (more readily reduced).

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "Consider the following 4 electrodes $\begin{aligned} &A: Ag^+(0.0001M)/Ag(s); \quad \parallel & B: Ag^+(0.1M)/Ag(s); \quad \parallel & C: Ag^+(0.01M)/Ag(s); \quad \parallel & \dots \end{aligned}$ " [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "For spontaneity of a cell, which is correct?..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "Consider the following electrodes $\begin{aligned} &P = Zn^{2+}(0.0001M)/Zn, Q = Zn^{2+}(0.1M)/Zn \quad \parallel & R = Zn^{2+}(0.01M)/Zn, S = Zn^{2+}(0.001M)/Zn \dots \end{aligned}$ " [Ans: N/A]

3.6 Kohlrausch's Law & Weak Electrolyte Dissociation (Moderate Yield — 5 PYQs)

Core Principle: Independent migration of ions at infinite dilution: $\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$. Degree of dissociation $\alpha = (\Lambda_m / \Lambda_m^\circ)$, and Ostwald's dilution law $K_a = (c\alpha^2 / 1 - \alpha)$.

■ **Caution / JEE Pitfall:** 1. Forgetting to multiply ionic conductances by stoichiometric coefficients (e.g., for $\text{Al}_2(\text{SO}_4)_3$ implies $2\lambda^\circ(\text{Al}^{3+}) + 3\lambda^\circ(\text{SO}_4^{2-})$). 2. For weak electrolytes, if $\alpha \ll 1$, do not approximate $1 - \alpha \approx 1$.

■ **Speed Trick:** Λ_m° for weak electrolytes like CH_3COOH can be obtained using strong electrolytes: $\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl})$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): " $\Lambda_m^\circ(\text{m}(\text{NH}_4\text{OH}))$ is equal to _____." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The value of 'A' in the equation $\lambda_m(m) = \lambda_m^\circ - A\sqrt{C}$ is same for the pair..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): " $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightleftharpoons 2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq})$ E_{cell}° at 25°C for the cell is 0.22V . The equilibrium constant K is _____." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The conductivity of centimolar solution of KCl at 298 K is $0.021\text{Ohm}^{-1}\text{cm}^{-1}$ and the resistance of the cell containing the solution at 298 K is 60Ω . The value of cell constant (G^*) is _____.

- (A) 3.28cm^{-1}
- (B) 1.26cm^{-1}
- (C) 3.34cm^{-1} [Correct]
- (D) 1.34cm^{-1}

Correct Answer: C

Solution Analysis: Conductivity is related to conductance and cell constant by $\kappa = G \times G^*$ where $G = (1 / R)$. So, $\kappa = (1 / R) \times G^*$. This gives $G^* = \kappa \times R$. Given, $\kappa = 0.021\text{Ohm}^{-1}\text{cm}^{-1}$ and $R = 60\Omega$. So, $G^* = 0.021 \times 60 = 1.26\text{cm}^{-1}$. So, the correct option is 1.26cm^{-1} Option B.

Benchmark Problem 2: KCET 2026 (Q#2)

Given below are the half-cell reactions: $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ ($E^\circ = -1.18\text{V}$) $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ ($E^\circ = +1.51\text{V}$) The E_{cell}° for $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ will be _____.

- (A) -2.69V , the reaction will not occur (Non-Spontaneous)
- (B) -2.69V , the reaction will occur (Spontaneous)
- (C) -0.33V , the reaction will not occur (Non-Spontaneous)
- (D) -0.33V , the reaction will occur (Spontaneous)

Correct Answer: None

Solution Analysis: For the reaction $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ one part of Mn^{2+} is getting reduced to Mn: $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ $E^\circ = -1.18\text{V}$. So this acts as the cathode reaction. The other part of Mn^{2+} is getting oxidised to Mn^{3+} : $2\text{Mn}^{2+} \rightarrow 2\text{Mn}^{3+} + 2\text{e}^-$. This is the reverse of $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ $E^\circ = +1.51\text{V}$. So the oxidation potential will be $E_{\text{ox}}^\circ = -1.51\text{V}$. Now, $E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ + E_{\text{anode}}^\circ$.

Benchmark Problem 3: KCET 2026 (Q#3)

$\Lambda_m(\text{NH}_4\text{OH})^0$ is equal to _____

- (A) $\Lambda_m(\text{NH}_4\text{OH})^0 + \Lambda_m(\text{NH}_4\text{Cl})^0 - \Lambda_m(\text{HCl})^0$
- (B) $\Lambda_m(\text{NH}_4\text{Cl})^0 + \Lambda_m(\text{NaOH})^0 - \Lambda_m(\text{NaCl})^0$
- (C) $\Lambda_m(\text{NH}_4\text{Cl})^0 + \Lambda_m(\text{NaCl})^0 - \Lambda_m(\text{NaOH})^0$
- (D) $\Lambda_m(\text{NaOH})^0 + \Lambda_m(\text{NaCl})^0 - \Lambda_m(\text{NH}_4\text{Cl})^0$

Correct Answer: None

Solution Analysis: For NH_4OH , $\Lambda_m(\text{NH}_4\text{OH})^0 = \lambda_{\text{NH}_4^+}^0 + \lambda_{\text{OH}^-}^0$ Using Kohlrausch's law, we combine strong electrolytes whose ions can give NH_4^+ and OH^- . Now, $\Lambda_m(\text{NH}_4\text{Cl})^0 = \lambda_{\text{NH}_4^+}^0 + \lambda_{\text{Cl}^-}^0$
 $\Lambda_m(\text{NaOH})^0 = \lambda_{\text{Na}^+}^0 + \lambda_{\text{OH}^-}^0$ $\Lambda_m(\text{NaCl})^0 = \lambda_{\text{Na}^+}^0 + \lambda_{\text{Cl}^-}^0$ So,
 $\Lambda_m(\text{NH}_4\text{OH})^0 = \Lambda_m(\text{NH}_4\text{Cl})^0 + \Lambda_m(\text{NaOH})^0 - \Lambda_m(\text{NaCl})^0$

Benchmark Problem 4: KCET 2026 (Q#4)

During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at cathode under STP conditions is

- (A) 22.4 L **[Correct]**
- (B) 11.2 L
- (C) 2.24 L
- (D) 1.12 L

Correct Answer: A

Solution Analysis: During electrolysis of acidified water, hydrogen gas is formed at the cathode and oxygen gas is formed at the anode. The overall reaction is $2\text{H}_2\text{O}(l) \rightarrow 2\text{H}_2(g) + \text{O}_2(g)$ From the equation, 1 mole of O_2 gives 2 moles of H_2 . Given mass of oxygen gas: 16g Molar mass of O_2 is 32g mol^{-1} So, moles of O_2 formed: $(16 / 32) = 0.5 \text{ mol}$ Since hydrogen is formed in double the number of moles, moles of $\text{H}_2 = 2 \times 0.5 = 1 \text{ mol}$ At STP, 1 mole of any gas occupies 22.4L So, volume of H_2 gas liberated is

Benchmark Problem 5: KCET 2019 (Q#2)

An aqueous solution of CuSO_4 is subjected to electrolysis using inert electrodes. The pH of the solution will

- (A) increase
- (B) remains unchanged
- (C) decrease
- (D) increase or decrease depending on the strength of the current

Correct Answer: None

Solution Analysis: In aqueous solution of CuSO_4 , the ions present in solution are Cu^{2+} , SO_4^{2-} , H^+ and OH^- . Possible reaction that can occur are as follows At Cathode $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(s)$, $E_{\text{Cu}^{2+}/\text{Cu}}^\circ = +0.34\text{V}$
 $2\text{H}_2\text{O}(l) + 2\text{e}^- \rightarrow \text{H}_2(g) + 2\text{OH}^-(\text{aq})$, $E_{\text{H}_2\text{O}/\text{H}_2}^\circ = -0.83\text{V}$ Due to high value of $E_{\text{Cu}^{2+}/\text{Cu}}^\circ$ Cu is released at cathode. At anode $2\text{H}_2\text{O}(l) \rightarrow \text{O}_2(g) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$, $E_{\text{H}_2\text{O}/\text{O}_2}^\circ = -1.23\text{V}$ & $2\text{SO}_4^{2-}(\text{aq}) \rightarrow \text{S}_2\text{O}_8^{2-} + 2\text{e}^-$